Project Roadmap – Project 3

MANET Routing in ns-3: AODV vs OLSR vs DSDV under Density and Mobility

Updated roadmap for the midterm checkpoint

Data Communications – Group Project, 2026–2027

Group Lupu Gabriel-Irinel, Filimon Alexandra-Elena, Ermilios Ilias-Alexandros
Repository `https://github.com/whitezxx/project_3_manet-ns3_unstpb_etti`
Environment ns-3.48 (optimized build), Ubuntu 24.04 on WSL2, Python 3 (pandas, numpy, matplotlib)

1 Summary

The midterm subset of the matrix (50 nodes, static, three protocols, one seed: 3 runs) is complete. It is reproduced from the CSV logs with one command (`run_midterm.sh`), which builds the program, runs the simulations and produces the PDR and delay plots. The run times measured in this phase are far below the initial estimate: a 50-node static run takes 5–6 s instead of 3 min, and a 100-node run about 33 s instead of 15 min. The budget for the full core matrix therefore drops from 6.2 h serial to well under one hour. **The committed scope is unchanged**: 75 runs at 5 seeds per point, plus Extensions B (energy) and C (bursty traffic).

2 Responsibilities and status

Member and part	Deliverable (initial roadmap)	Weeks	Status
Lupu Gabriel-Irinel	Scenario program: topology, Wi-Fi PHY/MAC, mobility, protocol selection, traffic, command-line parameters	2–6	**Done**
(simulation, C++)	Sanity tests	2–6	**Done**
	Extension B (energy)	10	**Planned**
Filimon Alexandra-Elena	Per-packet traces, control-packet counting, CSV log format	3–5	**Done**
(instrumentation, batch)	Parallel, resumable batch script for the matrix	8–10	**Planned**
	Repository and one-command reproduction	8–10	**Partial** (midterm only)
Ermilios Ilias-Alexandros	Python analysis: warm-up filtering, KPIs, figures from logs	4–7	**Done** (baseline)
(analysis, report)	95 % confidence intervals	4–7	**Planned**
	Report and slides; Extension C (bursty traffic)	11–14	**Planned**

The confidence intervals need at least 5 seeds per point, so they move to the matrix analysis (weeks 9–10). Every member commits from their own account; decisions and sanity checks are logged in `NOTES.md`.

3 Progress by phase

Phase	Status	Evidence
P0 Foundations	**Done**	ns-3.48 (optimized) built; reference example `manet-routing-compare` runs and its output was analysed (initial roadmap, Sec. 7).
P1 Scenario script	**Done**	`ns3/manet-compare.cc`: protocol, nodes, speed, pause, seed, flows, packet size and rate, range set from the command line. Per-packet CSV logs (tx, rx, control), FlowMonitor XML → CSV, per-run summary with wall-clock time. Sanity test with 2 nodes, 1 flow, 30 s: 56/56 packets delivered, delay ≈ 2 ms.
P2 Baseline	**Done**	50 nodes, static, AODV/OLSR/DSDV, seed 1. PDR and delay plots generated by `run_midterm.sh` from the CSV logs.
P3 Core matrix	**Planned**	Not started. Batch launch in week 8, as in the initial roadmap.
P4 Analysis, report	**Planned**	Weeks 9–14.

4 Midterm baseline result

50 nodes, static, seed 1, warm-up 20 s. Offered load: 10 CBR/UDP flows × 2 pkt/s × 128 B = 20 pkt/s (20.48 kbps). 200 s simulated, 180 s measured, 3580 packets offered per protocol.

Protocol	Sent	Received	PDR (%)	p50 (ms)	p95 (ms)	Jitter (ms)	NRL
AODV	3580	3217	89.9	1.92	2.32	0.51	4.14
OLSR	3580	3222	90.0	1.92	2.32	0.10	2.30
DSDV	3580	3148	87.9	1.92	3.12	8.46	6.34

Observations.

- Flow 2 (node 23 → node 1) delivers no packets under any protocol because node 1 is isolated (network partition). This caps PDR at ≈ 90 %. Excluding this flow: AODV 99.9 %, OLSR 96.4 % (whole run), DSDV 92.8 %. Partitions therefore already occur at 50 nodes, not only at 25 nodes as anticipated in the initial risk table.
- DSDV converges more slowly than the warm-up (whole run, flow 2 excluded): 181 losses in 0–20 s, 57 in 20–40 s, 10 in 40–60 s. OLSR loses packets only during 0–20 s.
- Control traffic: DSDV shows periodic peaks (≈ 15 s, full-table dumps); OLSR is flat (≈ 50 pkt/s, HELLO + TC); AODV runs at 50–100 pkt/s (HELLO + RREQ).
- With one seed, the AODV/OLSR PDR difference (0.1 pp) is not significant. Conclusions wait for 5 seeds with confidence intervals.

5 Changes with respect to the initial roadmap

Implemented

- **Traffic generator.** The CBR flows are produced by a custom sender (`CbrSend`) instead of `OnOffApplication`. With OnOff, OLSR offered 3431 packets against 3954 for AODV/DSDV, because OnOff stops generating packets while the source has no route. `CbrSend` offers exactly 2 pkt/s per flow under every protocol. Packets with no route at the source are counted as `src_drops`.
- **Fair comparison.** Fixed random-number streams (`AssignStreams`, `SetStream`) as planned. Verified on seed 1: the flow tables of the three protocols are identical (`diff` of `_flows.csv` is empty).
- **Control overhead.** Counted at the IPv4 `Tx` trace of every node, by UDP destination port (AODV 654, OLSR 698, DSDV 269), including originated and forwarded transmissions. Sanity check (seed 1): each log contains only the port of the selected protocol.
- **Run-time estimate** replaced by measured times (Section 6).

Planned (not yet implemented)

- **Hop count.** The initial roadmap took it from FlowMonitor. FlowMonitor's `avgHops` (already exported) ignores the warm-up and also counts forwards of packets that are lost later. It will be measured instead from the IP TTL at reception (`LocalDeliver` trace), after the warm-up.
- **Partition classification.** Node positions will be logged. Each flow is then classified as reachable or partitioned (BFS on the 100 m unit-disk graph), and PDR is reported both overall and over reachable flows.

No seed is discarded.

- **Warm-up.** 20 s remains the default. Because of the slow DSDV convergence, a 20 s vs 40 s sensitivity check on 5 seeds will decide the final value. DSDV convergence time will be reported as a separate result.

Cut

Nothing from the committed scope has been cut: the core matrix (75 runs, 5 seeds), all mandatory KPIs and both extensions (B, C) are kept. As in the initial roadmap, DSR, the Cooja/RPL variant and OMNeT++ remain out of scope. With the measured run times, the fallbacks planned for slow 100-node runs (overnight batches, 3 seeds at 100 nodes) are no longer expected to be needed.

6 Experiment matrix and updated run-time budget

The one-factor-at-a-time design of the initial roadmap is unchanged: density sweep {25, 50, 100} nodes at 0 m/s and speed sweep {1, 10} m/s at 50 nodes, with the 50-node static point shared; RngSeed = 12345, RngRun = 1, . . . , 5.

Nodes	Speed (m/s)	Neighbours	Initial estimate / run	Measured / run	Runs
25	0	3.1	1 min	not yet measured	15
50	0	6.3	3 min	5–6 s	15
50	1	6.3	3 min	not yet measured	15
50	10	6.3	3 min	≈ 20 s	15
100	0	12.6	15 min	≈ 33 s (OLSR only)	15
Serial total			**375 min**	**≈ 20 min**	**75**

The new total uses the measured values and conservative guesses for the unmeasured points (25 nodes: 3 s; 50 nodes at 1 m/s: 10 s): $15 \times (3 + 6 + 10 + 20 + 33)\,\text{s} = 1080\,\text{s} \approx 18\,\text{min}$ serial, about 5 min with 4 parallel runs. Even at three times these values, the matrix finishes in about one hour. AODV and DSDV at 100 nodes will be timed in week 8. Peak memory for a 50-node run is about 43 MB.

Reporting. Every KPI is given as mean ± 95 % confidence interval, $\bar{x} \pm t_{0.975,4}\, s/\sqrt{5}$ with $t_{0.975,4} = 2.776$, together with offered load, run duration and number of samples. Delay is reported as mean, p50 and p95; p99 only when a run has at least 1000 samples.

7 Plan to the final checkpoint

Week	Work item	Owner
7–8	Midterm presentation and demo.	all
8	Parallel, resumable batch script; timing of AODV/DSDV at 100 nodes; launch of the core matrix.	Filimon
8	Hop count from TTL; position log for partition classification.	Lupu
8–9	Warm-up sensitivity (20 s vs 40 s), final decision recorded in `NOTES.md`.	Ermilios
9–10	Matrix analysis with 95 % confidence intervals; partition classification; one-command reproduction of the full matrix.	Ermilios, Filimon
10	Extension B: energy (`WifiRadioEnergyModel`), joules per delivered packet per protocol.	Lupu
11–12	Extension C: bursty traffic (10 Hz for 3 s every 30 s). Analysis of why the results look as they do.	Ermilios
13–14	Report, slides, final presentation and live demo.	all

8 Risks and fallbacks – status at midterm

Risk	Status at midterm	Fallback
100-node runs slower than estimated	Largely resolved: 100 nodes (OLSR) take ≈ 33 s; AODV and DSDV still to be timed.	Unchanged: overnight runs on the fastest laptop; at least 3 seeds at 100 nodes with written justification.
Unfair comparison between protocols	Fixed streams implemented; identical flow tables verified on seed 1. The batch script will check every run.	–
Partitioned network	Observed already at 50 nodes (flow 2, seed 1).	Reported as a result, plus PDR over reachable flows; seeds are not discarded.
DSDV not converged within the warm-up (new)	Losses up to ≈ 60 s at 50 nodes.	Sensitivity check 20 s vs 40 s; convergence time reported separately.
Build or memory problems	No issue: optimized build, ≈ 43 MB per 50-node run.	One laptop used as the shared batch machine.
Schedule delay	On schedule for the week-8 batch launch.	Extensions are dropped, not the core matrix.